

PROJECT AND TEAM INFORMATION

Project Title

SkillArc

Student / Team Information

Team Name: Team #	Apex
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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

Problem Description

- **Lack of learning direction:** While learning a topic such as DSA, programming, AI, and ML, students often get confused about what and when to learn. With a variety of content available on the internet, it can often be overwhelming to find a proper course.
- **Lack of regular practice:** Students may find it difficult to practice coding questions regularly on platforms like LeetCode. Without enough motivation or a way to track their progress, they may slowly lose interest and stop practicing.
- **No reward for hardwork:** Students solve coding questions, keep streaks, make notes, and help their friends. But their hard work is not rewarded.
- **Progress tracking:** Students use different websites for coding, courses, notes, and practice. So it is hard for them to see all their progress in one place and the topics they have finished.
- **Healthy competition:** With no proper direction and practice, students may lose track of their own progress. Students need to be motivated and driven to keep a habit of practicing coding consistently. Leaderboards on the app can help students compete with themselves and others.
- **Accuracy of notes:** Since the app allows students to share their notes, there might not be a proper monitoring system for the accuracy of the notes uploaded. Sometimes notes can be misleading and not up to date with the latest information.

Importance

- **Practicing technical skills:** Students need to practice coding and problem solving so they can remember what they learned and not forget the topics for exam.
- **Roadmaps:** A roadmap shows students what to learn step by step. It helps them learn easily without looking at many different places.
- **Motivation and streaks:** Maintaining streaks can help to motivate students to practice coding on a daily basis. Credits, achievements, and progress bars can help to keep track of the students' progress.
- **Group learning and teaching:** Students make notes and teach each other. This helps them learn better.
- **Progress tracking and consistency:** Students can see their credits, streaks, achievements, and leaderboard. This helps them keep learning and coding.
- **Rewards:** Students can use their credit points at the end of each month to get vouchers.

State of the Art / Current solution (1 pt)

- CS students now use different websites and platforms to plan and manage their learning. Students find it hard to keep everything together.
- roadmap.sh gives students learning paths for subjects such as Computer Science fundamentals, C, C++ and Data Structures and many others.
- LeetCode is a favorite for students who want to practice coding. It gives Study Plans for topics like DSA, Dynamic Programming.
- Students also look to YouTube to learn concepts and to find extra study material.
- Because these resources are on sites students often have to switch from one website to another for learning, quizzes, coding practice, notes.
platform works well for a job but no single place on which students manage all their learning activities together.
- Our project SkillArc brings all these activities onto one platform. It will offer roadmaps, study materials, quizzes, coding practice, shared notes, weekly contests and leaderboards.
- The Credit Point system will give students learning activities, with their progress, achievements and rewards that students can earn.

Project Goals and Milestones (2 pts)

Project Goal

The goal of SkillArc is to develop a gamified student learning platform that provides structured, subject-wise and topic-wise roadmaps for DSA and C++. It will provide learning resources, notes, quizzes, and coding practice to help students maintain consistent learning. The platform will use Credit Points, weekly contests, and leaderboards to encourage regular participation and healthy competition among students. Students will also be able to share notes and other learning materials. Based on the Credit Points they earn, students can receive monthly rewards or vouchers, which will motivate them to develop productive learning habits while applying DSA and C++ OOP concepts in the project.

3-Month Project Milestones

Milestone	Technology / Area	Timeline
Requirement Analysis & UI/UX	Figma	Week 1–2
Frontend & Backend Infrastructure	HTML, CSS, JavaScript, Python/Flask, MySQL	Week 3–4
DSA Roadmap	DSA, LeetCode	Week 5
OOPs Roadmap	C++ / Java	Week 6
Frontend Roadmap	HTML, CSS, JavaScript	Week 7
Notes & Credit System	Backend, MySQL	Week 8–9
Streak, Leaderboard & Parent Module	JavaScript, Backend	Week 10–11
Integration, Testing & Deployment	GitHub, Hosting	Week 12

Project Approach (3 pts)

Part	Technology / Tech Stack	Description
DSA Implementation	C	Arrays, Linked Lists, Stacks, Queues, Graphs, Hashing, Searching and Sorting will be used for managing learning content, questions, student records, and leaderboard data.
C++ OOP Implementation	C++	Classes will be created for major entities such as Student, Subject, Topic, Quiz, Contest, Note, and Reward using OOP concepts and file handling.
Platform Interface	HTML, CSS, JavaScript, Python, Streamlit	The frontend and interactive interface will provide subject-wise roadmaps, study material, quizzes, coding practice, progress tracking, note sharing, contests, leaderboards, Credit Points, and rewards.
Data & Integration	SQLite, JSON, Git/GitHub	SQLite or JSON will be used for storing project data, while Git/GitHub will be used for managing and integrating the project code.

System Architecture (High Level Diagram)(2 pts)



Project Outcome / Deliverables (1 pts)

- **Subject-wise Roadmaps:** We believe that the subject wise roadmaps provide learning paths to students in different fields like DSA, programming. Subject wise roadmaps offer the option to categorize every topic in specific order so students can follow without getting confused.
- **Student Dashboard:** The Student Dashboard is our feature that provides an overview of completed topics coding activities, credits, achievements and overall progress. Student dashboard makes it convenient for students and teachers to monitor student's progress.
- **Credit-Based Reward System:** Credit Based Reward System is a feature where students can gain credits by solving coding problems keeping streaks creating and sharing notes and helping peers. Credit-Based Reward system is a way to acknowledge student's effort and reward them with credits for further incentive.
- **Streak and Achievement Tracking:** Streak and achievement tracking: Streak and Achievement tracking is a tool that tracks learning and coding streak as well as achievements to keep students motivated. Streak and achievement tracking helps students stay updated on the progress and practice regularly.
- **Notes and Peer-Learning Module:** Notes and Peer-Learning Module is a platform where students can add, share and access notes and learn together effectively. Notes and Peer Learning Module is a way to engage students in effective teamwork. Helps students learn from each other.
- **Progress Reports:** Progress reports is where we save the records of activities, credits, streaks and achievement to assess student performance. Progress Report is a way to evaluate the student progress and assist them if needed.

Assumptions

Students using skillarc shall have the basic computer knowledge which will make them use platform for learning, quizzes, coding practice and tracking their progress. Roadmaps, study materials will be provided through the platform. Students are expected to participate regularly and earn the credit point through solving the problems, maintaining streaks and sharing useful notes. These credits will be used to track progress and decide leaderboard positions and eligible rewards. Notes shared will be reviewed first before giving the contribution points. We assume that the required software, development tools and technologies will be available to the team for developing and testing the platform.

References

1. <https://roadmap.sh/roadmaps>
2. <https://takeuforward.org/home>
3. <https://www.geeksforgeeks.org/>
4. <https://www.youtube.com/>
5. Yashavant Kanetkar, Data Structures Through C, 5th Edition.
6. Herbert Schildt, C++: The Complete Reference, 4th Edition, McGraw-Hill.